

Weak lensing of the Lyman- α forest and its cross-correlation with CMB lensing: literature review and the sample-variance-limited estimator

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Abstract

These notes accompany an attempt to detect gravitational lensing of the DESI DR1 Lyman- α forest through its cross-correlation with CMB lensing maps from ACT and Planck. Section 1 reviews the (small) literature on lensing of the forest and the adjacent literature on forest–CMB-lensing correlations and on lensing reconstruction from three-dimensional source fields. Section 2 derives, in the flat-sky, single-source-plane, zero-noise (sample-variance) limit, the optimal quadratic estimator for the convergence from a set of forest pixels, the cubic estimator $W_{ijk}\delta_F^i\delta_F^j\delta_{F\kappa_{\text{CMB}}}^k$ for the cross-spectrum $C_L^{\kappa_{\text{Ly}\alpha}\kappa_{\text{CMB}}}$, its variance, the dense-sightline continuum limit with a closed-form reconstruction noise $N_\kappa(L)$, and a bias-hardened variant that projects out the response of the forest power spectrum to long-wavelength modes. Section 3 evaluates $N_\kappa(L)$ and the cross-correlation signal-to-noise for a fiducial forest power spectrum and DESI-like sightline densities. The central qualitative result is that for sparse sightlines most of the naive sample-variance information comes from an isotropic dilation of the transverse correlation function that is degenerate with a local modulation of the forest power spectrum amplitude, and that the robust (shear-like) information is set by the transverse rather than the line-of-sight resolution.

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1 Literature review

1.1 Lensing of the forest

Only one group has published on reconstructing the lensing potential from the forest itself. Croft et al. [1] proposed the idea: the three-dimensional forest is treated as a stack of source screens at $z \simeq 2\text{--}3$ and a configuration-space quadratic estimator, adapted from 21 cm lensing work, is applied to simulated spectra on a $15^\circ \times 15^\circ$ field with a sightline every 1.8 arcmin (which they call the extreme high end of achievable densities). They recover the degree-scale morphology of the potential and give no per-survey signal-to-noise. Metcalf et al. [2] supplied Gaussian noise forecasts. The noise per potential mode scales as $\sigma_\tau^2/[j_{\max}(l_{\max} - l_{\min})^2 l_{\min}^2]$, where σ_τ is the fractional per-pixel error and $j_{\max}$ the number of radial modes. Their cumulative signal-to-noise on the convergence power spectrum for DESI (770k spectra, $14\,000\text{ deg}^2$, 8.1 arcmin mean separation, per-pixel S/N 3.3) is 7 for $\Delta z = 0.1$ source bins and 29 for $\Delta z = 0.5$ bins, corresponding to a 1.5 per cent measurement of the amplitude of mass fluctuations at $z \simeq 2.5$; eBOSS gives 4–16 and an MSE-like survey 50–160. Metcalf et al. [3] replaced the harmonic estimator by a real-space minimum-variance pixel-pair estimator built on the full pixel covariance, with the potential expanded in Legendre polynomials over a patch. This handles irregular sightline positions, holes and inhomogeneous noise. On Gaussian mocks they find that high-fidelity maps require $\gtrsim 0.5$ sources per arcmin² and signal-dominated pixels; a density of 200 deg^{-2} fails. Shaw et al. [4] applied that estimator to hydrodynamic simulations with ray-traced potentials: forest non-Gaussianity reduces the signal-to-noise by a factor of about 2.7 in the noise-free case and 1.5 for unit signal-to-noise spectra, and the recovered potential is biased by 5–25 per cent, which they correct by Gaussianizing the flux field. Shaw et al. [5] forecast DESI directly: 50 quasars per deg^2 , 700k spectra, pixel noise $\sigma_\delta = 0.3$ after rebinning to $2.8\text{ \AA}$ and discarding the noisiest quarter, and detection defined as the correlation of the reconstructed potential with the foreground galaxy distribution. The full-survey significance is $S/N \simeq 3$ (two-pixel estimator), 4 (four-pixel) and 9 without pixel noise; $200\text{--}2000\text{ deg}^{-2}$ surveys on the same area reach 16–27. They state explicitly that they do not consider cross-correlation with CMB lensing.

No detection of forest lensing has been claimed, no analysis of eBOSS or DESI data exists, and no paper considers the cross-correlation of a forest-lensing reconstruction with CMB lensing. The last point is the gap this project targets: ACT DR6 [6, 7] and Planck PR4 [8] lensing maps cover the full DESI DR1 forest footprint of more than 9500 deg^2 with 428 403 valid Ly α -region forests [9], and a cross-correlation is immune to the additive Gaussian ($N^{(0)}$ -type) bias that dominates an auto-spectrum measurement.

1.2 Forest–CMB-lensing correlations that are not lensing of the forest

A separate line of work correlates the forest flux, or its local one-dimensional power, with κ_{CMB} . Vallinotto et al. [10, 11] forecast the two-point $\langle \delta_F \kappa_{\text{CMB}} \rangle$ and the $\langle \delta_F^2 \kappa_{\text{CMB}} \rangle$ correlations; Doux et al. [12] detected the latter at 5σ with BOSS DR12 and Planck 2015; Chiang and Slosar [13] modelled it as the squeezed gravitational bispectrum plus a term from mean-flux misestimation on finite skewers; Karaçaylı et al. [14] detected it at 4.8σ with DESI Y1 forests and Planck PR3; La Posta and Schaen [15] forecast $S/N \simeq 10, 15$ and 20 for DESI with ACT, Simons Observatory and CMB-S4. Physically these measure the response of the small-scale forest power to long-wavelength modes, quantified with separate-universe simulations by Chiang et al. [16]. This is important here for a negative reason: as shown in Section 2.3, the same response mimics a lensing convergence in a quadratic estimator, so the detected $P_F \times \kappa_{\text{CMB}}$ bispectrum is the

leading contaminant of the measurement we want to make. Loverde et al. [17] discuss the related but distinct effect of quasar magnification bias on forest statistics, at the 0.1–1 per cent level.

1.3 Lensing reconstruction from three-dimensional fields

The formalism below follows the CMB quadratic estimator of Hu [18], Hu and Okamoto [19] generalized to a three-dimensional source by Zahn and Zaldarriaga [20], who showed that each line-of-sight Fourier mode acts as an independent screen and that the screens combine with inverse-variance weights. Lu and Pen [21] pointed out that source non-Gaussianity sets a noise floor even without instrumental noise. Pourtsidou and Metcalf [22, 23] developed the 21 cm intensity-mapping version at $z \simeq 2\text{--}3$ including discreteness noise. Foreman et al. [24] computed the bias to the three-dimensional estimator from the matter bispectrum, found it to cost a factor of a few in signal-to-noise for 21 cm surveys, and introduced a bias-hardened estimator; Schaan et al. [25], Schaan and Ferraro [26] generalized the estimator to weakly non-Gaussian sources and introduced the shear-only estimator that is immune to isotropic modulations of the source power. Bias hardening in the CMB context is due to Namikawa et al. [27]; see also Sailer et al. [28], Maniyar et al. [29]. Buncher et al. [30] recently applied the standard and shear-only estimators to projected galaxy clustering as the source, the closest analogue of a sparse-tracer reconstruction. A fuller annotated list is kept in `report/literature_notes.md`.

2 Formalism in the sample-variance limit

2.1 Set-up and conventions

We work in the flat-sky, distant-observer approximation. A point in the forest volume is labelled by an angular position $\boldsymbol{\theta}$ (radians) and a comoving line-of-sight distance χ ; the effective comoving distance to the forest is χ_* ($\chi_* \simeq 3956 h^{-1}\text{Mpc}$ at $z = 2.4$ for a Planck-2018 cosmology). The flux contrast $\delta_F(\boldsymbol{\theta}, \chi)$ is treated as a statistically homogeneous Gaussian random field with the anisotropic (redshift-space) power spectrum $P_F(k_{\parallel}, k_{\perp})$. Its Fourier transform in the mixed angular/radial variables is

$$\delta(\boldsymbol{l}, k_{\parallel}) = \int d^2\theta d\chi \delta_F(\boldsymbol{\theta}, \chi) e^{-i(\boldsymbol{l}\cdot\boldsymbol{\theta} + k_{\parallel}\chi)}, \quad \langle \delta(\boldsymbol{l}, k_{\parallel}) \delta(\boldsymbol{l}', k'_{\parallel}) \rangle = (2\pi)^3 \delta_{\text{D}}^2(\boldsymbol{l} + \boldsymbol{l}') \delta_{\text{D}}(k_{\parallel} + k'_{\parallel}) \mathcal{P}(\boldsymbol{l}, k_{\parallel}), \quad (1)$$

with $\mathcal{P}(\boldsymbol{l}, k_{\parallel}) = P_F(k_{\parallel}, k_{\perp} = l/\chi_*)/\chi_*^2$. Reality of δ_F implies $\delta(-\boldsymbol{l}, -k_{\parallel}) = \delta^*(\boldsymbol{l}, k_{\parallel})$. For a survey of solid angle Ω and radial depth D the coincident delta functions are $(2\pi)^2 \delta_{\text{D}}^2(0) = \Omega$ and $2\pi \delta_{\text{D}}(0) = D$.

Lensing by a single effective source plane at χ_s is described by a projected potential $\phi(\boldsymbol{\theta})$ with deflection $\boldsymbol{\alpha} = \nabla\phi$, convergence $\kappa = -\nabla^2\phi/2$ (so $\kappa(\boldsymbol{L}) = L^2\phi(\boldsymbol{L})/2$) and shear $\gamma_1 = -(\partial_1^2 - \partial_2^2)\phi/2$, $\gamma_2 = -\partial_1\partial_2\phi$, exactly the convention of Hu and Okamoto [19]. The observed forest field is

$$\tilde{\delta}(\boldsymbol{\theta}, \chi) = \delta_F(\boldsymbol{\theta} + \nabla\phi(\boldsymbol{\theta}), \chi). \quad (2)$$

Because the source plane is common to all χ , the displacement is independent of the radial coordinate; this is the single approximation that makes the problem separable in $k_{\parallel}$. In reality the lensing kernel changes by a few per cent across a forest, which we defer to later work.

2.2 Mode coupling induced by lensing

Expanding (2) to first order, $\tilde{\delta} = \delta_F + \nabla\phi \cdot \nabla_{\boldsymbol{\theta}}\delta_F$, and Fourier transforming,

$$\tilde{\delta}(\boldsymbol{l}, k_{\parallel}) = \delta(\boldsymbol{l}, k_{\parallel}) - \int \frac{d^2l_1}{(2\pi)^2} [\boldsymbol{l}_1 \cdot (\boldsymbol{l} - \boldsymbol{l}_1)] \phi(\boldsymbol{l} - \boldsymbol{l}_1) \delta(\boldsymbol{l}_1, k_{\parallel}). \quad (3)$$

For a fixed realization of ϕ the two-point function of the observed field acquires an off-diagonal part. With $\mathbf{L} = \mathbf{l} + \mathbf{l}'$ and $\mathbf{L} \neq 0$,

$$\langle \tilde{\delta}(\mathbf{l}, k_{\parallel}) \tilde{\delta}(\mathbf{l}', k'_{\parallel}) \rangle_{\phi} = 2\pi \delta_D(k_{\parallel} + k'_{\parallel}) \left[(2\pi)^2 \delta_D^2(\mathbf{L}) \mathcal{P}(l, k_{\parallel}) + f(\mathbf{l}, \mathbf{l}'; k_{\parallel}) \phi(\mathbf{L}) \right], \quad f(\mathbf{l}, \mathbf{l}'; k_{\parallel}) = \mathbf{L} \cdot \mathbf{l} \mathcal{P}(l, k_{\parallel}) + \mathbf{L} \cdot \mathbf{l}' \mathcal{P}(l') \quad (4)$$

It is convenient to express the response in terms of the convergence, $f_{\kappa} \equiv 2f/L^2$, so that the off-diagonal term reads $f_{\kappa} \kappa(\mathbf{L})$. Equation (4) is diagonal in $k_{\parallel}$: every radial Fourier mode is an independently lensed two-dimensional screen with power spectrum $\mathcal{P}(l, k_{\parallel})$, all sharing the same ϕ [20]. Pairs with $k'_{\parallel} \neq -k_{\parallel}$ carry no lensing information and, for a Gaussian field, no covariance either, so an optimal estimator never uses them.

The pair-displacement picture. The same result follows from the real-space statement in the task description: a pair of pixels $i = (a, p)$ and $j = (b, q)$ on sightlines a, b at radial positions χ_p, χ_q has covariance

$$C_{ij}(\phi) = \xi(\boldsymbol{\theta}_{ab} + \boldsymbol{\alpha}_a - \boldsymbol{\alpha}_b, \chi_p - \chi_q) \simeq \xi(\boldsymbol{\theta}_{ab}, \Delta\chi_{pq}) + \nabla_{\theta} \xi(\boldsymbol{\theta}_{ab}, \Delta\chi_{pq}) \cdot [\nabla \phi(\boldsymbol{\theta}_a) - \nabla \phi(\boldsymbol{\theta}_b)], \quad (5)$$

where ξ is the correlation function of δ_F in angular-radial coordinates and $\boldsymbol{\theta}_{ab} = \boldsymbol{\theta}_a - \boldsymbol{\theta}_b$. Only the relative deflection of the two members of a pair matters, so a pair measures the local tidal field of ϕ rather than the deflection itself. Fourier transforming (5) over both positions reproduces (4); this was verified symbolically for a periodic one-dimensional toy model in `mathematica/pair_displacement_check.wls`.

2.3 Squeezed limit: dilation, shear and the amplitude degeneracy

For lens modes much larger than the forest modes, $L \ll l$, expanding (4) to second order in L gives (verified in `mathematica/lensing_response_checks.wls`, check A)

$$f_{\kappa}(\mathbf{l}, \mathbf{L} - \mathbf{l}) \simeq 2\mathcal{P}(l) + 2(\hat{\mathbf{L}} \cdot \hat{\mathbf{l}})^2 l \mathcal{P}'(l) = \underbrace{[2\mathcal{P} + l\mathcal{P}']}_{\text{dilation}} + \underbrace{l\mathcal{P}' \cos 2\varphi_{lL}}_{\text{shear}}, \quad (6)$$

with $\mathcal{P}' = \partial\mathcal{P}/\partial l$ at fixed $k_{\parallel}$. The two pieces have a clean geometric meaning. For a pair separation θ_{ab} small compared with the lens wavelength, $\boldsymbol{\alpha}_a - \boldsymbol{\alpha}_b \simeq (\boldsymbol{\theta}_{ab} \cdot \nabla) \nabla \phi = -(\kappa \mathcal{K} + \gamma) \boldsymbol{\theta}_{ab}$, so the pair correlation is evaluated at the true separation $(1 - \kappa) \boldsymbol{\theta}_{ab} - \gamma \boldsymbol{\theta}_{ab}$: the convergence isotropically dilates the transverse correlation function while the shear makes it anisotropic. In Fourier space the exact power spectrum of a field dilated and sheared by constant $\kappa, \gamma_1, \gamma_2$ is $\mathcal{P}(|M^{-T} \mathbf{l}|) / \det M$ with $M = \mathcal{K} - \kappa \mathcal{K} - \gamma$, whose first-order expansion equals (6) (check B in the same script).

The amplitude degeneracy is now apparent. A slowly varying multiplicative modulation of the field, $\tilde{\delta} = (1 + a) \delta_F$, produces the coupling

$$\langle \tilde{\delta}(\mathbf{l}, k_{\parallel}) \tilde{\delta}(\mathbf{l}', k'_{\parallel}) \rangle \supset 2\pi \delta_D(k_{\parallel} + k'_{\parallel}) f_a(\mathbf{l}, \mathbf{l}'; k_{\parallel}) a(\mathbf{L}), \quad f_a = \mathcal{P}(l, k_{\parallel}) + \mathcal{P}(l', k_{\parallel}) \xrightarrow{L \ll l} 2\mathcal{P}, \quad (7)$$

so the dilation term $2\mathcal{P} + l\mathcal{P}' = (2 + n)\mathcal{P}$, with $n = d \ln \mathcal{P} / d \ln l$, is proportional to the amplitude response. For a transverse spectrum that is locally a power law the convergence is therefore *indistinguishable* from a local change of the forest power spectrum amplitude by $\delta \ln P_F = (2 + n)\kappa$; only the shear term $l\mathcal{P}' \cos 2\varphi$ and departures from a power law break the degeneracy. For $n = -2$ the dilation response vanishes identically, and for a spectrum that is flat in l ($n = 0$, which is what the forest looks like transversely whenever $l/\chi_* \ll k_{\parallel}$) there is no shear response at all and the entire lensing signal is an amplitude change. This matters because the forest power spectrum really is modulated by long-wavelength density and tidal modes, with responses measured by Chiang et al. [16], and those long modes are correlated with κ_{CMB} . That correlation is exactly the $P_F \times \kappa_{\text{CMB}}$ bispectrum detected by Doux et al. [12], Karaçaylı et al. [14], and it will enter our cubic estimator as an additive bias unless projected out (Section 2.7).

2.4 The optimal quadratic estimator for discrete pixels

Let $d_i = \tilde{\delta}(\boldsymbol{\theta}_{a(i)}, \chi_{p(i)})$, $i = 1 \dots N_{\text{pix}}$, be the vector of all forest pixels on all sightlines, in the zero-noise limit. Expand the potential in a set of real basis functions, $\phi(\boldsymbol{\theta}) = \sum_{\mu} p_{\mu} \psi_{\mu}(\boldsymbol{\theta})$; in the continuum limit below the ψ_{μ} are Fourier modes, while on a finite irregular footprint one may use Legendre or Zernike polynomials as in Metcalf et al. [3]. To first order in p the pixel covariance is, from (5),

$$C_{ij}(p) = C_{ij}^0 + \sum_{\mu} p_{\mu} C_{ij}^{\mu}, \quad C_{ij}^{\mu} = \nabla_{\theta} \xi(\boldsymbol{\theta}_{ab}, \Delta \chi_{pq}) \cdot [\nabla \psi_{\mu}(\boldsymbol{\theta}_a) - \nabla \psi_{\mu}(\boldsymbol{\theta}_b)]. \quad (8)$$

For a Gaussian likelihood $-2 \ln \mathcal{L} = d^T C^{-1} d + \ln \det C$ the Newton–Raphson step from $p = 0$ is the optimal (minimum-variance, unbiased to first order) quadratic estimator,

$$\hat{p}_{\mu} = \frac{1}{2} \sum_{\nu} F_{\mu\nu}^{-1} \left[d^T C_0^{-1} C^{\nu} C_0^{-1} d - \text{Tr}(C_0^{-1} C^{\nu}) \right], \quad F_{\mu\nu} = \frac{1}{2} \text{Tr}(C_0^{-1} C^{\mu} C_0^{-1} C^{\nu}). \quad (9)$$

Writing $\hat{p}_{\mu} = \sum_{ij} E_{ij}^{\mu} d_i d_j - b_{\mu}$ identifies the quadratic weights and the mean field,

$$E^{\mu} = \frac{1}{2} \sum_{\nu} F_{\mu\nu}^{-1} C_0^{-1} C^{\nu} C_0^{-1}, \quad b_{\mu} = \sum_{ij} E_{ij}^{\mu} C_{ij}^0. \quad (10)$$

Using $\langle d_i d_j \rangle = C_{ij}(p)$ one checks $\langle \hat{p}_{\mu} \rangle = p_{\mu} + O(p^2)$, and for Gaussian d the covariance at $p = 0$ is $\text{Cov}(\hat{p}_{\mu}, \hat{p}_{\nu}) = F_{\mu\nu}^{-1}$, which saturates the Cramér–Rao bound. The convergence coefficients follow by applying $-\nabla^2/2$ to the basis; for Fourier modes $\hat{\kappa}_{\mathbf{L}} = (L^2/2)\hat{\phi}_{\mathbf{L}}$. Two remarks. First, the same formulae hold with pixel noise by replacing $C^0 \rightarrow C^0 + N$, so nothing in this section is specific to the sample-variance limit except the numerical value of F . Second, because E^{μ} is built from ratios of C^0 and its derivative, an overall rescaling of the forest power spectrum leaves the estimator and its variance unchanged: in the zero-noise limit the reconstruction noise depends only on the *shape* of P_F and on the geometry of the pixel set.

2.5 The cubic cross-correlation estimator

Let m_k be the pixels of a CMB convergence map and $\kappa_{\mathbf{L}}^{\text{CMB}} = \sum_k Y_{\mathbf{L}k} m_k$ its (masked, apodized) flat-sky harmonic transform, normalized so that $\langle \kappa_{\mathbf{L}} \kappa_{\mathbf{L}'}^* \rangle = \Omega \delta_{\mathbf{L}\mathbf{L}'} C_{\mathbf{L}}$ for a full-sky-normalized spectrum. Choosing Fourier modes as the basis in (9), the natural bandpower estimator for the cross-spectrum in a bin b containing N_b modes is

$$\hat{C}_b = \frac{1}{\Omega N_b} \sum_{\mathbf{L} \in b} \text{Re}[\hat{\kappa}_{\mathbf{L}} \kappa_{\mathbf{L}}^{\text{CMB}*}] = \sum_{ijk} W_{ijk}^b d_i d_j m_k - \frac{1}{\Omega N_b} \sum_{\mathbf{L} \in b} \text{Re}[b_{\mathbf{L}} \kappa_{\mathbf{L}}^{\text{CMB}*}], \quad W_{ijk}^b = \frac{1}{\Omega N_b} \sum_{\mathbf{L} \in b} \text{Re}[E_{ij}^{\mathbf{L}} Y_{\mathbf{L}k}^*]. \quad (11)$$

This is the cubic form $W_{ijk} \delta_F^i \delta_F^j \kappa_{\text{CMB}}^k$ anticipated in the task description, with the weights fixed by (10). Its properties:

- **Expectation.** To first order in ϕ , $\langle \hat{C}_b \rangle = \bar{C}_b^{\kappa_{\text{Ly}\alpha} \kappa_{\text{CMB}}}$, the bin average of the true cross-spectrum. The mean-field term is a fixed function of the pixel geometry and is uncorrelated with κ_{CMB} , so it averages to zero but should be subtracted to avoid excess variance.
- **No Gaussian ($N^{(0)}$) bias.** The disconnected part of $\langle d_i d_j m_k \rangle$ vanishes because $\langle m_k \rangle = 0$ and the Gaussian part of $d_i d_j$ is independent of m_k . This is the main advantage of the cross-correlation over the auto-spectrum of $\hat{\kappa}$, whose $N^{(0)}$ bias equals the full reconstruction noise.
- **Second-order lensing ($N^{(1)}$ -type) bias.** At $O(\phi^2)$, contracting κ_{CMB} with one power of ϕ and the remaining ϕ with the two δ_F fields gives a term proportional to $\int d^2 L' C_{L'}^{\kappa_{\text{Ly}\alpha} \kappa_{\text{CMB}}} \times$ (coupling kernel), the analogue of the CMB $N^{(1)}$ bias. It is a few-per-cent correction that must be computed with the fiducial spectra.

- **Intrinsic clustering bias.** δ_F is itself a biased tracer of the matter at $z \simeq 2-3$, which lies inside the CMB lensing kernel. The connected bispectrum $\langle \delta_F \delta_F \kappa_{\text{CMB}} \rangle_c$ therefore contributes even in the absence of lensing of the forest. Its squeezed part is the response of P_F to long modes, $\delta \ln P_F = R_\delta \delta_L + R_K K_L$ (density and tidal responses), correlated with κ_{CMB} ; in the language of Section 2.3 it enters as $\langle \hat{a}_{\mathbf{L}} \kappa_{\mathbf{L}}^{\text{CMB}*} \rangle \neq 0$ projected onto $\hat{\kappa}$ through the response ratio $F_{\kappa\alpha}/F_{\kappa\kappa}$. This is the forest analogue of the gravitational-nonlinearity bias of Foreman et al. [24] and it is not small: it is the signal that Karaçaylı et al. [14] detected at 4.8σ in DESI Y1.
- **Variance.** For Gaussian fields and optimal weights the variance in the continuum limit is

$$\text{Var}(\hat{C}_b) = \frac{(C_L^{\kappa_{\text{Ly}\alpha} \kappa_{\text{Ly}\alpha}} + N_\kappa(L))(C_L^{\kappa_{\text{CMB}} \kappa_{\text{CMB}}} + N_L^{\text{CMB}}) + (C_L^{\kappa_{\text{Ly}\alpha} \kappa_{\text{CMB}}})^2}{f_{\text{sky}}(2L+1)\Delta L}, \quad (12)$$

with $N_\kappa(L)$ the reconstruction noise derived next. For a general pixel set the numerator's first factor is replaced by the appropriate combination of F^{-1} , i.e. by $\text{Cov}(\hat{\kappa}_{\mathbf{L}}, \hat{\kappa}_{\mathbf{L}'})$ from (9).

2.6 Continuum limit: Hu–Okamoto form and reconstruction noise

For infinitely dense sightlines and complete radial coverage over a depth D , the field is measured on a regular three-dimensional grid and the Fisher matrix of Fourier modes becomes diagonal. It is then simpler to derive the estimator directly. Take

$$\hat{\kappa}(\mathbf{L}) = A(L) \int \frac{dk_{\parallel}}{2\pi} \int \frac{d^2l}{(2\pi)^2} g(\mathbf{l}, \mathbf{L} - \mathbf{l}; k_{\parallel}) \tilde{\delta}(\mathbf{l}, k_{\parallel}) \tilde{\delta}(\mathbf{L} - \mathbf{l}, -k_{\parallel}). \quad (13)$$

Inserting (4), the coincident radial delta function gives $2\pi\delta_{\text{D}}(0) = D$, so $\langle \hat{\kappa}(\mathbf{L}) \rangle = \kappa(\mathbf{L})$ requires

$$A(L)^{-1} = D \int \frac{dk_{\parallel}}{2\pi} \int \frac{d^2l}{(2\pi)^2} g f_{\kappa}. \quad (14)$$

The Gaussian variance of (13) at $\phi = 0$ has two Wick pairings, each producing another factor D and the power spectra of the two legs; minimizing it subject to the normalization gives the standard weight $g = f_{\kappa}/[2\mathcal{P}(l, k_{\parallel})\mathcal{P}(|\mathbf{L} - \mathbf{l}|, k_{\parallel})]$ and

$$\boxed{\frac{1}{N_{\kappa}(L)} = D \int_{-\infty}^{\infty} \frac{dk_{\parallel}}{2\pi} I_{\kappa\kappa}(L, k_{\parallel}), \quad I_{XY}(L, k_{\parallel}) = \int \frac{d^2l}{(2\pi)^2} \frac{f_X(\mathbf{l}, \mathbf{L} - \mathbf{l}; k_{\parallel}) f_Y(\mathbf{l}, \mathbf{L} - \mathbf{l}; k_{\parallel})}{2\mathcal{P}(l, k_{\parallel}) \mathcal{P}(|\mathbf{L} - \mathbf{l}|, k_{\parallel})}} \quad (15)$$

with $\langle \hat{\kappa}(\mathbf{L}) \hat{\kappa}^*(\mathbf{L}') \rangle = (2\pi)^2 \delta_{\text{D}}^2(\mathbf{L} - \mathbf{L}') [C_L^{\kappa\kappa} + N_{\kappa}(L)]$. Equation (15) is the inverse-variance sum over independent $k_{\parallel}$ screens of the Hu–Okamoto noise of each screen, $N_{k_{\parallel}}^{-1} = I_{\kappa\kappa}$, with the mode density $D dk_{\parallel}/2\pi$ counting the screens [20]. It is the same object as F^{-1} of (9) per unit solid angle. Two consistency checks: (i) since $f_{\kappa} \propto \mathcal{P}$, the integrand is invariant under $\mathcal{P} \rightarrow c\mathcal{P}$, as it must be in the sample-variance limit; (ii) for a white spectrum, $\mathcal{P} = \text{const}$, $f_{\kappa} = 2\mathcal{P}$ exactly, so $I_{\kappa\kappa} = l_{\text{max}}^2/2\pi$ and $N_{\kappa}^{-1} = D k_{\text{max}} l_{\text{max}}^2/2\pi^2$; this equals twice the number of real degrees of freedom per steradian of the field, which is what one obtains directly by noting that a dilation by κ changes the variance of every independent white-noise pixel by a factor $(1+2\kappa)$ and each pixel then carries Fisher information $\frac{1}{2}(2)^2 = 2$. The numerical implementation (`code/recon_noise.py`) reproduces this count to the accuracy of the $|\mathbf{L} - \mathbf{l}| \leq l_{\text{max}}$ edge exclusion.

The cutoffs l_{max} and $k_{\parallel, \text{max}}$ are where the real survey enters even in this idealization: transverse modes are only measured up to the sightline Nyquist frequency, $l_{\text{max}} \simeq \pi/\bar{\theta}_{\text{sep}} = \pi\sqrt{n_q}$ for n_q sightlines per steradian, radial modes up to a fraction of the pixel Nyquist frequency, and radial modes below $k_{\parallel, \text{min}} \sim 0.03 h \text{ Mpc}^{-1}$ are removed by continuum fitting. With sightlines of mean forest length D_f covering only part of the total depth, the product $D l_{\text{max}}^2$ that controls (15) is

fixed by the total sightline length per unit area, so one may use $D = D_f$ together with the full n_q . Sparse sampling also aliases small-scale transverse power into the measured band; that effect is beyond the continuum treatment and is one reason the numbers below are optimistic.

2.7 Hardening against amplitude modulation

An estimator $\hat{a}(\mathbf{L})$ for the amplitude modulation is built from (13) with f_a in place of f_κ , and the two estimators respond to each other's field with response functions $R_{\kappa a}(L) = F_{\kappa a}/F_{\kappa\kappa}$ and $R_{a\kappa} = F_{\kappa a}/F_{aa}$, where $F_{XY} = D \int dk_{\parallel}/2\pi I_{XY}$. The bias-hardened estimator [27]

$$\hat{\kappa}^{\text{BH}} = \frac{\hat{\kappa} - R_{\kappa a}\hat{a}}{1 - R_{\kappa a}R_{a\kappa}}, \quad N_{\kappa}^{\text{BH}}(L) = \frac{N_{\kappa}(L)}{1 - \rho_{\kappa a}^2(L)}, \quad \rho_{\kappa a}^2 = \frac{F_{\kappa a}^2}{F_{\kappa\kappa}F_{aa}}, \quad (16)$$

has zero response to any $a(\boldsymbol{\theta})$ that multiplies all $k_{\parallel}$ modes equally. Since the true responses $R_{\delta}(k_{\parallel}, k_{\perp})$ of the forest are scale dependent, a more conservative choice hardens each $k_{\parallel}$ screen separately, $1/N_{\kappa}^{\text{BH, slice}} = D \int dk_{\parallel}/2\pi I_{\kappa\kappa}(1 - \rho_{\kappa a}^2(L, k_{\parallel}))$; this removes any modulation of the form $a(\boldsymbol{\theta}, k_{\parallel})$ and keeps only the information that is genuinely anisotropic or shape-changing within a screen. Both variants are evaluated below; the truth for the real forest lies in between and also requires hardening against the tidal response, which mimics shear rather than convergence and would need the $k_{\perp}$ -dependence of R_K to be separated from $l\mathcal{P}'$.

2.8 The signal and its detectability

In the Limber approximation, with a single forest source plane at χ_* and the CMB at χ_{CMB} ,

$$C_L^{XY} = \int_0^{\chi_{\min}} d\chi \frac{W_X(\chi)W_Y(\chi)}{\chi^2} P_{\text{m}}\left(k = \frac{L+1/2}{\chi}, z(\chi)\right), \quad W_s(\chi) = \frac{3}{2}\Omega_{\text{m}}\left(\frac{H_0}{c}\right)^2 (1+z)\chi \frac{\chi_s - \chi}{\chi_s}, \quad (17)$$

and the total detection significance of the cross-spectrum from (12) is

$$\left(\frac{S}{N}\right)^2 = \sum_L f_{\text{sky}}(2L+1) \frac{(C_L^{\kappa_{\text{Ly}\alpha}\kappa_{\text{CMB}}})^2}{(C_L^{\kappa_{\text{Ly}\alpha}\kappa_{\text{Ly}\alpha}} + N_{\kappa}(L))(C_L^{\kappa_{\text{CMB}}\kappa_{\text{CMB}}} + N_L^{\text{CMB}}) + (C_L^{\kappa_{\text{Ly}\alpha}\kappa_{\text{CMB}}})^2}. \quad (18)$$

Because the forest source plane sits well inside the CMB lensing kernel the two convergences are strongly correlated, with $r_L = C_L^{\kappa_{\text{Ly}\alpha}\kappa_{\text{CMB}}}/(C_L^{\kappa_{\text{Ly}\alpha}\kappa_{\text{Ly}\alpha}}C_L^{\kappa_{\text{CMB}}\kappa_{\text{CMB}}})^{1/2} \simeq 0.9$ at $L = 10$ falling to 0.67 at $L \gtrsim 300$.

3 Toy numbers

Fiducial model. The forest power spectrum is the Kaiser form with the non-linear correction of Arinyo-i-Prats et al. [31], $P_F = b_F^2(1 + \beta_F\mu^2)^2 P_{\text{lin}}(k) D_{\text{NL}}(k, \mu)$, with approximate $z = 2.4$ parameters $b_F = -0.13$, $\beta_F = 1.6$, $q_1 = 0.6$, $q_2 = 0$, $k_v = 1 h \text{ Mpc}^{-1}$, $a_v = 0.55$, $b_v = 1.6$, $k_p = 16 h \text{ Mpc}^{-1}$ (`code/forest_power.py`; only the shape matters here). The linear and non-linear matter power spectra come from CAMB for a Planck-2018 cosmology. The radial depth is $D = 350 h^{-1} \text{ Mpc}$ (one forest), the radial band is $0.03 \leq k_{\parallel} \leq k_{\parallel, \text{max}}$ with $k_{\parallel, \text{max}} \in \{1, 2, 5\} h \text{ Mpc}^{-1}$, and the transverse cutoff is the sightline Nyquist multipole for $n_q \in \{20, 50, 100, 400\} \text{ deg}^{-2}$ ($l_{\text{max}} = 1273$ for the DESI-DR1-like value $n_q = 50$; DR1 has about 45 forests per deg^2 over 9500 deg^2). Figure 1 shows the fiducial spectrum.

CMB lensing noise. Rather than fabricate noise curves, we use white noise levels calibrated to reproduce the published detection significances of the auto-spectra: $N_L^{\text{CMB}} = 1.68 \times 10^{-7}$ reproduces 43σ over $40 \leq L \leq 763$ on $f_{\text{sky}} = 0.23$ for ACT DR6 [6], and $N_L^{\text{CMB}} = 3.03 \times 10^{-7}$ reproduces the $\simeq 42\sigma$ -equivalent amplitude error over $8 \leq L \leq 400$ on $f_{\text{sky}} = 0.67$ for Planck

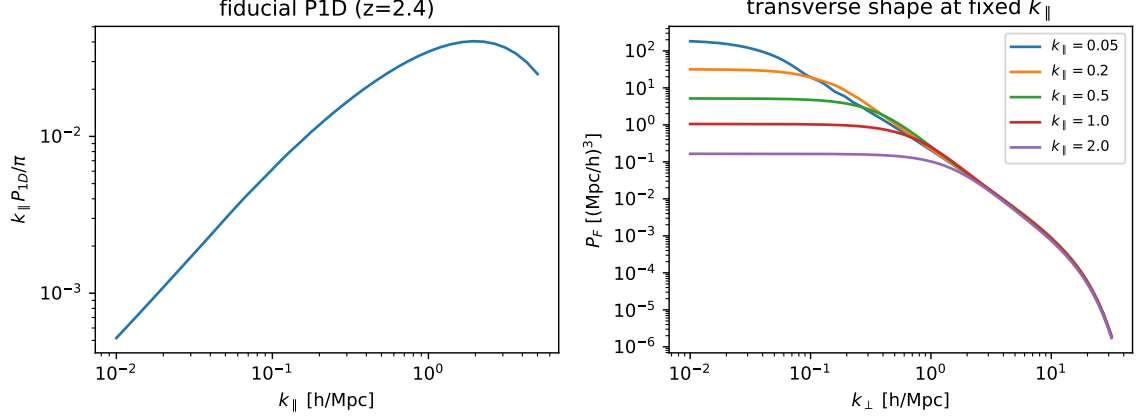


Figure 1: Fiducial forest power spectrum at $z = 2.4$. Left: the implied one-dimensional power. Right: the transverse shape at fixed $k_{\parallel}$; for $k_{\perp} \ll k_{\parallel}$ the spectrum is flat in l , which is the regime in which lensing is degenerate with an amplitude change (Section 2.3).

Table 1: Reconstruction noise at $L = 100$ and cumulative cross-correlation signal-to-noise (to $L = 3000$) in the sample-variance limit. N^{BH} is the per-screen amplitude-hardened noise. S/N columns are for $k_{\parallel, \text{max}} = 2$; the last column uses N^{BH} with ACT-like CMB noise.

n_q [deg $^{-2}$]	l_{max}	$k_{\parallel, \text{max}}$	$N_{\kappa}(100)$	$N_{\kappa}^{\text{BH}}(100)$	S/N ACT	S/N Planck	S/N ACT, BH
20	805	1.0	1.36×10^{-7}	9.49×10^{-7}	25.7	25.4	7.1
		2.0	5.71×10^{-8}	9.44×10^{-7}			
		5.0	2.04×10^{-8}	9.43×10^{-7}			
50	1273	1.0	6.21×10^{-8}	2.01×10^{-7}	36.5	35.7	15.2
		2.0	2.45×10^{-8}	1.98×10^{-7}			
		5.0	8.27×10^{-9}	1.97×10^{-7}			
100	1800	1.0	3.40×10^{-8}	6.63×10^{-8}	46.1	44.6	25.0
		2.0	1.33×10^{-8}	6.37×10^{-8}			
		5.0	4.26×10^{-9}	6.32×10^{-8}			
400	3600	1.0	8.15×10^{-9}	8.84×10^{-9}	67.4	64.2	54.4
		2.0	4.01×10^{-9}	7.33×10^{-9}			
		5.0	1.21×10^{-9}	7.05×10^{-9}			

PR4 [8]. Real noise curves rise steeply at high L and should replace these before any forecast is quoted externally. Overlap fractions of $f_{\text{sky}} = 0.15$ (ACT) and 0.22 (Planck, the full DESI DR1 forest area) are assumed.

Results. Figure 2 compares $N_{\kappa}(L)$ with the signal spectra and Table 1 lists values at $L = 100$, where $C_{100}^{\kappa_{\text{Ly}\alpha}\kappa_{\text{Ly}\alpha}} = 2.78 \times 10^{-8}$, $C_{100}^{\kappa_{\text{Ly}\alpha}\kappa_{\text{CMB}}} = 4.57 \times 10^{-8}$ and $C_{100}^{\kappa_{\text{CMB}}\kappa_{\text{CMB}}} = 1.38 \times 10^{-7}$.

Three features of these numbers are robust even though their absolute values are not:

1. For a DESI-like density the naive sample-variance noise at $L \sim 100$, $N_{\kappa} = 2.45 \times 10^{-8}$, is comparable to the signal $C_L^{\kappa_{\text{Ly}\alpha}\kappa_{\text{Ly}\alpha}}$, and it falls steeply with $k_{\parallel, \text{max}}$ (as $1/k_{\parallel, \text{max}}$ when the transverse spectrum is flat) and with n_q (as $1/l_{\text{max}}^2 \propto 1/n_q$).
2. The amplitude-hardened noise is ~ 8 times larger for $n_q = 50$ ($N_{\kappa}^{\text{BH}} = 1.98 \times 10^{-7}$), ~ 17 times larger for $n_q = 20$, and it is almost independent of $k_{\parallel, \text{max}}$. This is the quantitative form of Section 2.3: radial modes with $k_{\parallel} \gg l_{\text{max}}/\chi_*$ see a transversely flat spectrum, and for them lensing is purely an amplitude change. The information that survives hardening comes from the few screens with $k_{\parallel} \lesssim l_{\text{max}}/\chi_* \simeq 0.3 h \text{ Mpc}^{-1}$, so it is set by the sightline

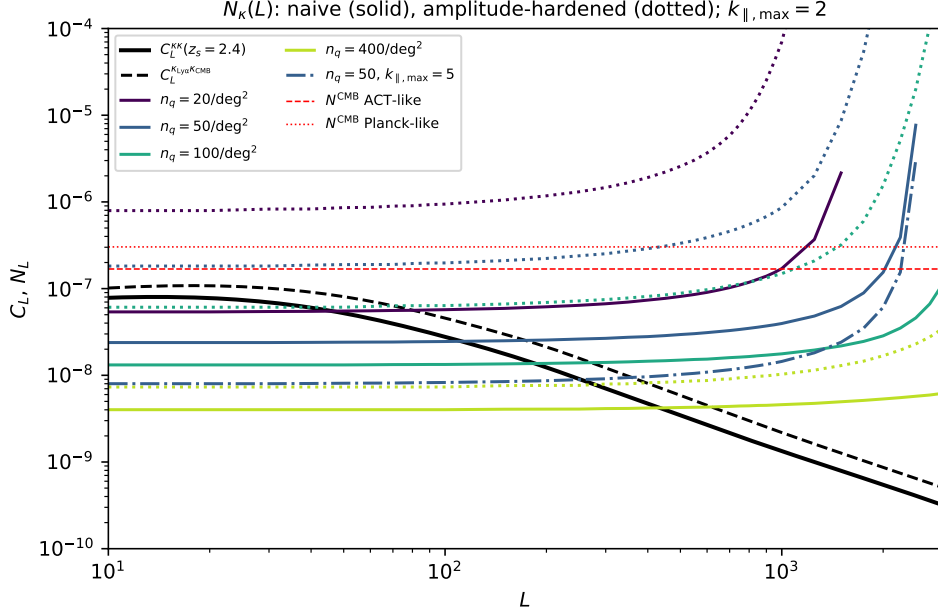


Figure 2: Sample-variance-limited reconstruction noise $N_\kappa(L)$ (solid, Eq. 15) for sightline densities n_q and $k_{\parallel, \text{max}} = 2 h \text{ Mpc}^{-1}$, its per-screen amplitude-hardened version (dotted, Eq. 16), and the case $k_{\parallel, \text{max}} = 5$ for $n_q = 50$ (dash-dotted). Black curves are the forest convergence auto-spectrum and its cross-spectrum with κ_{CMB} ; red lines are the white-noise equivalents of the ACT DR6 and Planck PR4 reconstruction noise.

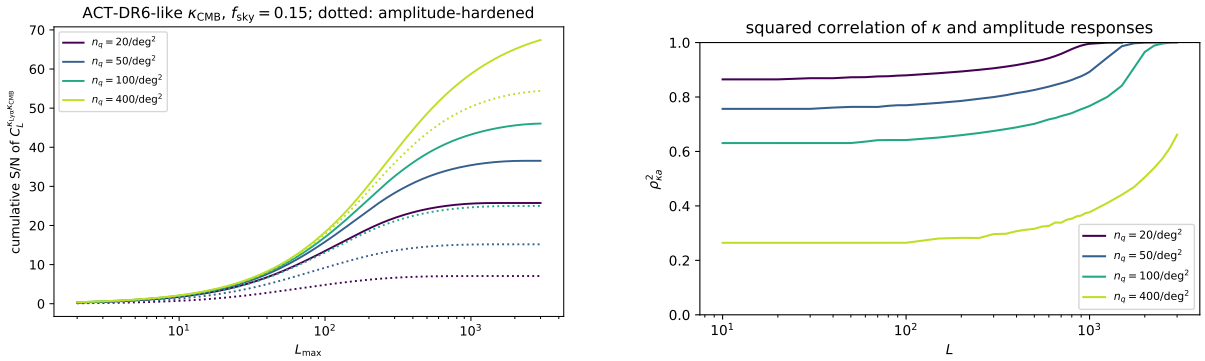


Figure 3: Left: cumulative signal-to-noise of $C_L^{\kappa_{\text{Ly}\alpha} \kappa_{\text{CMB}}}$ for ACT-like CMB noise and $f_{\text{sky}} = 0.15$, with (dotted) and without (solid) amplitude hardening. Right: squared correlation coefficient between the convergence and amplitude responses, $\rho_{\kappa a}^2$ of Eq. (16), after summing over $k_{\parallel}$ screens.

density alone. Increasing the density from 50 to 400 deg^{-2} reduces N_κ^{BH} by a factor ~ 30 , faster than the n_q^{-1} scaling of the naive noise, because it also converts amplitude-degenerate modes into shear-sensitive ones.

3. With ACT-like CMB noise the sample-variance-limited cross-correlation would be detected at $\simeq 37\sigma$ for $n_q = 50$ (Planck: 36σ), dominated by $L \lesssim 500$. These numbers are upper limits of no practical relevance by themselves: DESI pixel noise ($\sigma_\delta \simeq 0.8$ per $0.8 \text{ \AA}$ pixel against a forest rms of 0.34) removes most radial modes above $k_\parallel \sim 0.5 h \text{ Mpc}^{-1}$, sparse sampling aliases transverse power, and the estimator's own $N^{(1)}$ and clustering biases are not included. Shaw et al. [5] find $S/N \simeq 3\text{--}4$ for the reconstructed potential against foreground galaxies with realistic noise. The route from the numbers here to a realistic forecast is to add the pixel noise to $\mathcal{P}$ in (15), treat the sightline sampling explicitly through (9), and compute the bias terms.

4 Three tracers: separating lensing from the forest response

Section 3 showed that hardening the estimator against an amplitude modulation costs most of the sample-variance information. This section asks a different question: with three fields on the same footprint, the quasars q at the forest redshift, the forest δ_F and κ_{CMB} , which power spectra and bispectra separate the lensing of the forest from its response to long-wavelength density and tidal modes without assuming that response is known? The answer is formulated in the squeezed limit and at the level of spectra only.

4.1 Squeezed-limit model

Let $\hat{P}_F(\mathbf{k}|\boldsymbol{\theta})$ be the forest power spectrum measured locally in a cell around $\boldsymbol{\theta}$ over the redshift slab of interest (in practice the pixel-pair estimator of Section 2.4 restricted to a cell, or the position-dependent power spectrum). Its fractional fluctuation, $p(\mathbf{k};\boldsymbol{\theta}) = \delta \ln \hat{P}_F(\mathbf{k}|\boldsymbol{\theta})$, is a two-dimensional field for every short-mode bin $\mathbf{k} = (k_\parallel, k_\perp, \varphi)$, with φ the angle of $\mathbf{k}_\perp$ to the long-mode direction $\hat{\mathbf{L}}$. To first order in the long modes,

$$p(\mathbf{k}; \mathbf{L}) = T(\mathbf{k}) \kappa_{\text{Ly}\alpha}(\mathbf{L}) + \mathcal{R}(\mathbf{k}) \delta_s(\mathbf{L}) + \dots, \quad (19)$$

where δ_s is the matter density projected over the slab and:

- $T(\mathbf{k}) = (2 + n_\perp) + n_\perp \cos 2\varphi$, with $n_\perp = \partial \ln P_F / \partial \ln k_\perp$ at fixed $k_\parallel$, is the lensing template of Eq. (6). It is *known*, because it is built from the measured P_F itself.
- $\mathcal{R}(\mathbf{k}) = R_\delta(k_\parallel, k_\perp) + R_K(k_\parallel, k_\perp) \cos 2\varphi$ is the response of the forest to the long mode. For a long mode with $k_\parallel \simeq 0$, which is the only kind that projects onto a two-dimensional field such as κ_{CMB} , the tidal tensor is fixed by the density, $K_{ij} = (\hat{L}_i \hat{L}_j - \delta_{ij}/3) \delta_s$, so the tidal response is not an independent field: it is a $\cos 2\varphi$ harmonic multiplying the same $\delta_s(\mathbf{L})$ (`mathematica/three_tracer_checks.wls`, check 5). $\mathcal{R}$ is treated as *unknown*: one number per short-mode bin and harmonic.

Both terms have the same angular structure (an $m = 0$ and an $m = 2$ harmonic), so they cannot be separated by the angular dependence alone. They can be separated because they multiply different long-mode fields: $\kappa_{\text{Ly}\alpha}$ is sourced by matter at $z < z_s$, δ_s by matter at $z \simeq z_s$, and the quasars trace the latter and not the former.

The quasar overdensity projected over the same slab is

$$q(\mathbf{L}) = b_q \delta_s(\mathbf{L}) + m \kappa_q(\mathbf{L}) + \epsilon(\mathbf{L}), \quad m \equiv 5s - 2, \quad (20)$$

with $b_q \simeq 3.5$ the quasar bias, $m \kappa_q$ the magnification-bias term ($\kappa_q \simeq \kappa_{\text{Ly}\alpha}$ since the quasars sit at the forest redshift) and ϵ shot noise plus any stochasticity.

4.2 The measurements and the deprojection identity

Cross-correlating p with the two long-mode fields gives two families of bispectra (they are $\langle \delta_F \delta_F X \rangle$ in squeezed configurations, with the short modes binned in $\mathbf{k}$):

$$B_{FF\kappa}(\mathbf{k}; L) \equiv \langle p(\mathbf{k}) \kappa_{\text{CMB}} \rangle_L = \mathcal{R}(\mathbf{k}) C_L^{\delta_s \kappa_{\text{CMB}}} + T(\mathbf{k}) C_L^{\kappa_{\text{Ly}\alpha} \kappa_{\text{CMB}}}, \quad (21)$$

$$B_{FFq}(\mathbf{k}; L) \equiv \langle p(\mathbf{k}) q \rangle_L = \mathcal{R}(\mathbf{k}) C_L^{\delta_s q} + T(\mathbf{k}) C_L^{\kappa_{\text{Ly}\alpha} q}. \quad (22)$$

B_{FFq} is the quasar–forest–power cross-correlation proposed by Chiang et al. [16]; $B_{FF\kappa}$ integrated over $\mathbf{k}_\perp$ is the $P_{1D}-\kappa_{\text{CMB}}$ bispectrum of Doux et al. [12], Karaçaylı et al. [14], for which T vanishes identically because a single skewer is invariant under a transverse remapping (the dilation response integrates to a boundary term, $\int_0^{l_{\text{max}}} (2\mathcal{P} + l\mathcal{P}') l dl = l_{\text{max}}^2 \mathcal{P}(l_{\text{max}})$, check 4). Form the linear combination

$$\mathcal{D}(\mathbf{k}; L) \equiv B_{FF\kappa}(\mathbf{k}; L) - \beta_L B_{FFq}(\mathbf{k}; L) = T(\mathbf{k}) \left[C_L^{\kappa_{\text{Ly}\alpha} \kappa_{\text{CMB}}} - \beta_L C_L^{\kappa_{\text{Ly}\alpha} q} \right], \quad \beta_L \equiv \frac{C_L^{\delta_s \kappa_{\text{CMB}}}}{C_L^{\delta_s q}} \quad (23)$$

in which the unknown response cancels identically for every $\mathbf{k}$ (check 1). Equivalently, one cross-correlates the full quadratic estimator $\hat{\kappa}$ of Section 2.6 with the deprojected CMB map $\kappa_\perp \equiv \kappa_{\text{CMB}} - \beta q$. Three properties make this clean:

1. **No knowledge of the forest response is needed**, neither R_δ nor R_K , nor of the forest’s non-linear bispectrum: anything linear in the long mode cancels. The tidal part cancels with the same β_L because it multiplies the same δ_s .
2. **β_L is fixed by geometry**. In the Limber approximation with a slab kernel, $C_L^{\delta_s \kappa_{\text{CMB}}} = \overline{W_{\text{CMB}}} P / \chi_*^2$ and $C_L^{\delta_s q} = b_q P / (\chi_*^2 \Delta\chi)$, so $\beta_L = \overline{W_{\text{CMB}}} \Delta\chi / b_q$ up to slab-weighting corrections: the matter power spectrum drops out. Numerically $\beta = 0.084$ for the slab $2.1 < z < 3.0$ ($\Delta\chi = 712 h^{-1} \text{Mpc}$), constant in L to better than one per cent. It requires b_q (measured to a few per cent from C_L^{qq} or the three-dimensional quasar power spectrum) and Ω_m and distances. Alternatively β_L is measured directly as $C_L^{q\kappa_{\text{CMB}}} / (C_L^{qq} - 1/\bar{n}_q)$, which needs no bias at all.
3. **The residual is second order in imperfections**. If q has stochasticity $C^{\epsilon\epsilon}$ beyond Poisson noise, the residual response leak is $(1 - r_{q\delta}^2)$ of the original contamination, with $r_{q\delta}$ the deterministic quasar–matter correlation coefficient (check 2); Poisson noise in q adds variance but no bias. Ignoring the magnification term biases β by $\Delta\beta/\beta = m (C^{\delta_s \delta_s} C^{\kappa_q \kappa_{\text{CMB}}} - 2C^{\delta_s \kappa_{\text{CMB}}} C^{\delta_s \kappa_q}) / (b_q C^{\delta_s \delta_s} C^{\delta_s \kappa_{\text{CMB}}})$ (check 3b), which is 9 per cent at $L = 100$ for $m = 0.5$; since the contamination is a few times the signal (below), m must be known to ~ 20 per cent. It is measured by the cross-correlation of the quasars with a low-redshift galaxy sample g , $C_L^{qq} = m C_L^{\kappa_q g}$, which has no intrinsic overlap. The magnification term also feeds $C_L^{\kappa_{\text{Ly}\alpha} q} = m C_L^{\kappa_{\text{Ly}\alpha} \kappa_q}$ in (23), but multiplied by $\beta \simeq 0.08$ it changes the lensing signal by ~ 1.5 per cent.

The complete set of measurements is therefore: (i) $B_{FF\kappa}(\mathbf{k}; L)$, (ii) $B_{FFq}(\mathbf{k}; L)$, (iii) $C_L^{q\kappa_{\text{CMB}}}$, (iv) C_L^{qq} with shot noise subtracted, (v) C_L^{qg} with a low- z galaxy sample for magnification, and (vi) the forest power spectrum $P_F(\mathbf{k})$ that defines the template T . The lensing detection is the amplitude of $\mathcal{D}(\mathbf{k}; L)$ against the template $T(\mathbf{k})$; the internal consistency test is that $\mathcal{D}(\mathbf{k}; L)/T(\mathbf{k})$ is independent of $\mathbf{k}$, including the sign relation between the $m = 0$ and $m = 2$ harmonics, whereas any residual response would show up as a $\mathbf{k}$ -dependent residual. As a by-product, B_{FFq} divided by $C_L^{\delta_s q}$ measures $\mathcal{R}(\mathbf{k})$ itself, up to the small lensing leak $T m C^{\kappa_{\text{Ly}\alpha} \kappa_q} / (b_q C^{\delta_s \delta_s})$ of a few per cent.

Two practical caveats. The sightlines are quasars, and quasars in the template slab also provide forest pixels in the lower part of the same slab, so the local sightline density is correlated

Table 2: Reconstruction noise and cross-correlation signal-to-noise with realistic forest noise, for the slab $2.1 < z < 3.0$, $R_\delta = 2$, $b_q = 3.5$, $m = 0.5$, ACT-like CMB noise with $f_{\text{sky}} = 0.15$ (Planck-like with 0.22 in the last column). “Naive” is $\langle \hat{\kappa} \kappa_{\text{CMB}} \rangle$, whose expectation is biased by the amount in the fifth column; “BH” are the hardened estimators of Section 2.7; “deproj.” is Eq. (23).

n_{eff} [deg ⁻²]	P_N [h ⁻¹ Mpc]	$N_\kappa(100)$	$N_\kappa^{\text{BH}}(100)$	bias/signal at $L = 100$	naive	BH slice	cumulative S/N to $L = 3000$		
							BH global	deproj. ACT	deproj.
25	0.00	1.94×10^{-6}	4.92×10^{-6}	3.12	5.4	3.2	3.6	4.6	
25	0.17	5.36×10^{-6}	1.22×10^{-5}	3.12	3.2	2.1	2.2	2.8	
25	0.33	9.63×10^{-6}	2.16×10^{-5}	3.14	2.4	1.6	1.7	2.1	
50	0.00	5.94×10^{-7}	1.33×10^{-6}	2.93	9.8	6.4	7.1	8.4	
50	0.17	1.64×10^{-6}	3.22×10^{-6}	2.87	6.0	4.1	4.4	5.1	
50	0.33	2.89×10^{-6}	5.55×10^{-6}	2.89	4.5	3.2	3.3	3.8	
100	0.00	1.90×10^{-7}	3.87×10^{-7}	2.75	17.0	11.8	13.1	14.4	
100	0.17	5.32×10^{-7}	9.29×10^{-7}	2.62	10.5	7.8	8.4	9.0	
100	0.33	9.28×10^{-7}	1.58×10^{-6}	2.63	8.0	6.0	6.4	6.8	

with q ; any estimator bias that depends on the local density (aliasing or noise-bias subtraction errors) would leak into B_{FFq} . The clean solution is to take the template from quasars in the slab and the pixels from quasars behind it, which the forest geometry nearly enforces already ($z_{\text{pix}} \leq z_q - 0.04$). Second, magnification of the sightline quasars themselves correlates the sightline density with foreground structure, hence with both κ_{CMB} and $\kappa_{\text{Ly}\alpha}$; this is a mean-field effect, controlled by computing the estimator with the actual sightline positions.

4.3 Realistic noise

The sparse, noisy sampling of the forest enters (15) by replacing $\mathcal{P} \rightarrow \mathcal{P} + \mathcal{N}$ in the denominators, with the response f_κ unchanged (self-pairs of a sightline carry no lensing response, Eq. 8 with $a = b$, and only add variance). For Poisson-distributed sightlines with effective density $\bar{n}_{\text{eff}}$ per steradian at the redshift in question, the noise is white in l and given by the one-dimensional power of a skewer,

$$\mathcal{N}(k_\parallel) = \frac{P_N + P_{1D}(k_\parallel)}{\bar{n}_{\text{eff}}}, \quad P_N = \sigma_\delta^2 \Delta\chi_{\text{pix}}, \quad (24)$$

the form used for three-dimensional forest power spectra by McQuinn & White. The P_{1D} term is the aliasing of unresolved transverse power; it is present even for perfect spectra and dominates at DESI densities. For DESI DR1, $\sigma_\delta \simeq 0.8$ per $0.8 \text{ \AA}$ pixel ($\Delta\chi_{\text{pix}} \simeq 0.55 h^{-1}\text{Mpc}$ at $z = 2.4$) gives $P_N \simeq 0.33 h^{-1}\text{Mpc}$, to be compared with $P_{1D}(k_\parallel = 0.1) \simeq 0.19 h^{-1}\text{Mpc}$; the $P_N \simeq 0.17$ case corresponds to the cleaned sample of Shaw et al. [5]. About 45 forests per deg² each covering $\sim 350 h^{-1}\text{Mpc}$ of a $712 h^{-1}\text{Mpc}$ deep slab give $\bar{n}_{\text{eff}} \simeq 25 \text{ deg}^{-2}$ at any one redshift; 50 and 100 deg⁻² stand for a complete DESI or a denser survey. The quasar template has $\bar{n}_q = 25 \text{ deg}^{-2}$ in the slab (shot noise $1/\bar{n}_q = 1.2 \times 10^{-5} \text{ sr}$), the CMB noise is the white-noise equivalent of Section 3, and $\kappa_\perp = \kappa_{\text{CMB}} - \beta q$ has power $C^{\kappa_{\text{CMB}}\kappa_{\text{CMB}}}(1 - r_{q\kappa}^2) + N^{\text{CMB}} + \beta^2(C^{qq} + 1/\bar{n}_q)$ with $r_{q\kappa}^2 = 0.038$ at $L = 100$: deprojecting the quasars removes almost none of the CMB lensing signal, and the added shot-noise term $\beta^2/\bar{n}_q = 8.60 \times 10^{-8}$ is comparable to $C_{100}^{\kappa_{\text{CMB}}\kappa_{\text{CMB}}} = 1.38 \times 10^{-7}$.

4.4 Results

Table 2 and Figures 4–5 give the outcome. Three points:

1. **The contamination is large.** With $R_\delta = 2$ the response term in $\langle \hat{\kappa} \kappa_{\text{CMB}} \rangle$ is 1.64 times the lensing signal at $L = 40$ and 3.25 times at $L = 100$ (the ratio $C_L^{\delta\kappa_{\text{CMB}}}/C_L^{\kappa_{\text{Ly}\alpha}\kappa_{\text{CMB}}}$ grows with L because the slab density spectrum is shallower than the lensing spectrum), and the

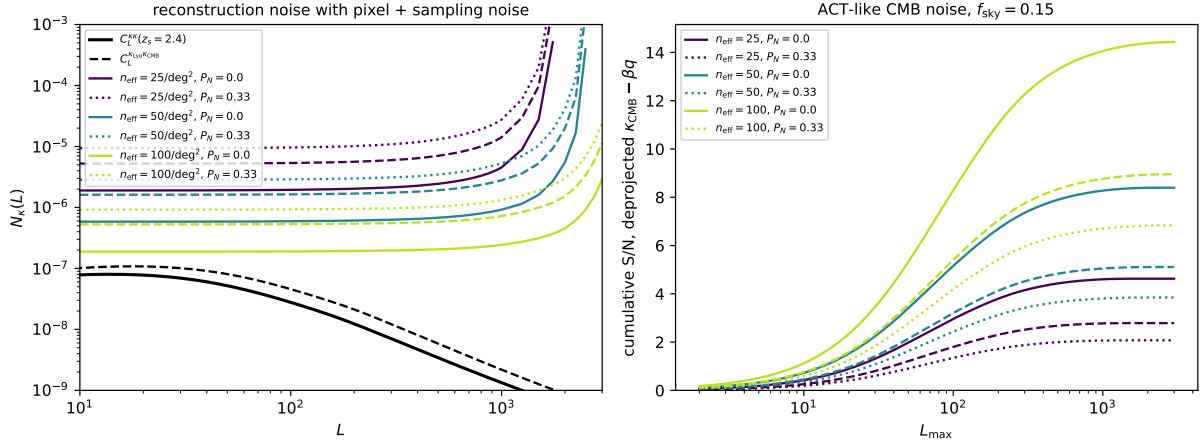


Figure 4: Left: reconstruction noise with the sampling and pixel noise of Eq. (24) (solid: perfect spectra, dotted: DESI DR1 pixel noise) for effective sightline densities of 25, 50 and 100 deg^{-2} . Right: cumulative signal-to-noise of the deprojected cross-correlation $\langle \hat{\kappa} (\kappa_{\text{CMB}} - \beta q) \rangle$.

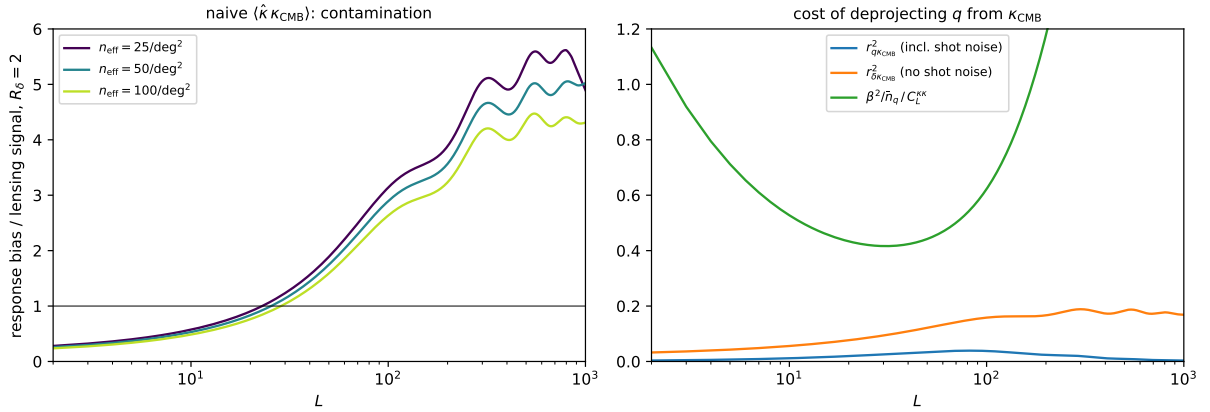


Figure 5: Left: ratio of the response contamination to the lensing signal in the naive $\langle \hat{\kappa} \kappa_{\text{CMB}} \rangle$ for a fiducial response $R_\delta = 2$, with DR1 noise. Right: the cost of deprojecting q from κ_{CMB} : the fraction of CMB lensing power removed ($r_{q\kappa}^2$) and the added quasar shot-noise term relative to $C_L^{\kappa_{\text{CMB}} \kappa_{\text{CMB}}}$.

estimator transfers it almost fully, $R_{\kappa a} \simeq 0.8$ –1. A naive measurement would be a $\sim 7\sigma$ “detection” dominated by the Karaçaylı et al. bispectrum. The deprojection is not optional.

2. **Sampling noise, not pixel noise, sets the scale.** Aliasing alone raises $N_\kappa(100)$ from 2.5×10^{-8} (dense sampling, Section 3) to 1.94×10^{-6} at 25 deg^{-2} ; DR1 pixel noise raises it further to 9.63×10^{-6} . Only modes with $k \lesssim 0.15 h \text{ Mpc}^{-1}$ are signal dominated.
3. **What remains.** For DR1-like parameters the deprojected cross-correlation reaches $S/N = 2.1$ with ACT-like and 2.1 with Planck-like CMB noise, against 2.4 for the (biased) naive statistic and 1.6 for the hardened one: deprojection costs ~ 15 per cent, hardening ~ 35 per cent. Cleaner spectra ($P_N = 0.17$) give 2.8; perfect spectra 4.6; a complete DESI with $\bar{n}_{\text{eff}} = 50$ and clean spectra 5.1. These are consistent with the $S/N \simeq 3$ –4 that Shaw et al. [5] find for the full DESI footprint against foreground galaxies, once scaled to their area. The CMB choice matters little because the forest side dominates the variance.

The conclusion for DESI DR1 is a marginal, $\sim 2\sigma$, clean measurement, with the response term itself detectable at high significance in the same data. Reaching 5σ requires an effective sightline density of $\sim 50 \text{ deg}^{-2}$ at fixed redshift with cleaned spectra, i.e. the complete DESI survey or DESI-II. Low-redshift galaxies as the lensing tracer avoid the deprojection but not the sampling noise, which is the actual limitation, so they offer no gain in signal-to-noise for the same forest data; their value is as an independent systematic check. Not included here, and all in the direction of lowering these numbers, are forest non-Gaussianity (a factor 1.5–2.7 in Shaw et al. 4), continuum-fitting distortions, and the fact that the fiducial noise level applies to the whole slab rather than the noisier high-redshift end.

5 Caveats and next steps

- **Single source plane.** Pixels at different χ are lensed by slightly different convergences; the estimator as written recovers a $k_\parallel$ -weighted average. Modelling this is straightforward within (9) (each screen gets its own W_s).
- **Gaussianity.** The forest is strongly non-Gaussian on the scales that carry the information. Shaw et al. [4] find a factor 1.5–2.7 loss in signal-to-noise; the variance (12) will need the forest trispectrum or simulations.
- **Clustering bias.** The dominant systematic is the intrinsic $\langle \delta_F \delta_F \kappa_{\text{CMB}} \rangle$ bispectrum. The plan is to compute its projection onto $\hat{\kappa}$ using the measured responses of Chiang et al. [16] and the DESI Y1 measurement of Karaçaylı et al. [14], and to design a hardened estimator that is orthogonal to both the density and tidal responses; the price is the factor ~ 8 in noise found above for the density part alone.
- **Data-level effects.** Continuum fitting removes low- $k_\parallel$ modes and distorts the measured ξ ; DLAs, metals and quasar magnification bias [17] add non-lensing structure; the CMB maps carry tSZ/CIB residuals correlated with the forest’s foregrounds.
- **Numerics to firm up.** Replace the white CMB noise by the public ACT DR6 and Planck PR4 noise curves; refine the Arinyo parameters to the published $z = 2.4$ values; add pixel noise and a realistic $n(z)$ of forests.
- **Pending adversarial review.** The derivations in Sections 2.2–2.6 and the numbers in Section 3 are queued for review with codex/gpt-6-astra, which was unavailable when they were written.

A Symbolic checks

The scripts in `mathematica/` verify: (A) the second-order squeezed expansion (6), its angular average $2\mathcal{P} + l\mathcal{P}'$, its $\cos 2\varphi$ component $l\mathcal{P}'$, and the power-law result $(2+n)\mathcal{P}$; (B) that the exact power spectrum of a field dilated and sheared by constant $\kappa, \gamma_1, \gamma_2$ agrees with (6) to first order, and that $\kappa = -\nabla^2\phi/2$ for the potential used; (C) that the real-space pair-displacement covariance (5), Fourier transformed in a periodic one-dimensional toy model, reproduces the mode coupling (4) for all tested (L, l) and vanishes for $l + l' \neq L$. `code/recon_noise.py` contains the white-spectrum mode-counting check of (15).

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